

## Jesse Liu

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| APPOINTMENTS             | Assistant Professor of Physics, New York University, USA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2025 – present |
|                          | Junior Research Fellow, Trinity College, University of Cambridge, UK                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2021 – 2024    |
|                          | Grainger Postdoctoral Fellow, University of Chicago, USA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2019 – 2021    |
| EDUCATION                | DPhil Particle Physics, University of Oxford, UK                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2015 – 2019    |
|                          | Thesis: ‘Discovery strategies for dark matter and Higgsinos at the LHC’                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                |
|                          | Supervisors: Alan Barr and John March-Russell                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                |
|                          | MSc Physics, Perimeter Institute and University of Waterloo, Canada                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2014 – 2015    |
|                          | Supervisor: Natalia Toro                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                |
|                          | BA Physics, University of Oxford, UK                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2011 – 2014    |
| AWARDS &<br>SCHOLARSHIPS | Idea Day Prize, ATLAS Collaboration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2021           |
|                          | Junior Research Fellowship (4 years, deferred to Oct 2021), Trinity College, Cambridge                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2019           |
|                          | Grainger Postdoctoral Fellowship (2 years, any field of physics), University of Chicago                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2019           |
|                          | Best Poster (1st out of 226 entries), International Conference on High Energy Physics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2018           |
|                          | Travel Grant, International Conference on High Energy Physics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2018           |
|                          | Santander Academic Travel Award, University of Oxford                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2018           |
|                          | STFC Postgraduate Studentship (3.5 years PhD scholarship)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2015           |
|                          | PSI Scholarship, Perimeter Institute and University of Waterloo                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2014           |
|                          | Fitzgerald Prize, Exeter College, Oxford                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2014           |
|                          | Summer Research Studentship, Oxford Particle Physics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2013           |
|                          | Gibbs Prize, winner of Oxford physics speaking competition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2013           |
|                          | East Scholarship, Exeter College, Oxford                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2012           |
|                          | Fitzgerald Prize, Exeter College, Oxford                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2012           |
|                          | Essay Prize, winner of Institute of Physics essay competition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2012           |
| PAPERS                   | All papers available at INSPIREHEP. ATLAS author since 2017. Particle physics convention lists authors alphabetically. Selected papers with direct personal contributions:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                |
|                          | <ul style="list-style-type: none"><li>• ATLAS Collaboration (<b>J. Liu</b> Editor &amp; Lead Analyzer), ‘Measurement of charged-particle production in <math>\sqrt{s_{\text{NN}}} = 9.62</math> TeV proton-oxygen collisions as a probe of cosmic-ray air showers with the ATLAS detector’, arXiv:2604.05512<br/>▷ <i>ATLAS Physics Briefing, NYU EPP News feature</i></li><li>• N. Fomin, B. Hommels, T. Ivison, K. Kariyapperuma and <b>J. Liu</b>, ‘Extended thermal cycling of ATLAS ITk strip modules with and without stress mitigating interposers’, JINST <b>20</b> (2025) P09036, arXiv:2507.12586<br/>▷ <i>NYU EPP News feature</i></li><li>• L. Beresford, S. Clawson and <b>J. Liu</b>, ‘Strategy to measure tau <math>g - 2</math> via photon fusion in LHC proton collisions’, Phys. Rev. D <b>110</b> (2024) 092016, arXiv:2403.06336</li><li>• S. Knirck, G. Hoshino, et. al. (BREAD Collaboration), ‘First Results from a Broadband Search for Dark Photon Dark Matter in the 44 to 52 <math>\mu\text{eV}</math> Range with a Coaxial Dish Antenna’, Phys. Rev. Lett. <b>132</b> (2024) 131004, arXiv:2310.13891<br/>▷ <i>PRL Editors’ Suggestion, UChicago News feature</i></li></ul> |                |

- ATLAS Collaboration (**J. Liu** Editor), ‘Observation of the  $\gamma\gamma \rightarrow \tau\tau$  Process in Pb+Pb Collisions and Constraints on the  $\tau$ -lepton Anomalous Magnetic Moment with the ATLAS Detector’, Phys. Rev. Lett. **131** (2023) 151802, arXiv:2204.13478  
▷ *PRL Editors’ Suggestion, ATLAS Physics Briefing, The Conversation feature*
- ATLAS Collaboration (**J. Liu** Chair of Editorial Board), ‘Exclusive dielectron production in ultraperipheral Pb+Pb collisions at  $\sqrt{s_{NN}} = 5.02$  TeV with ATLAS’, JHEP **06** (2023) 182, arXiv:2207.12781
- ATLAS Collaboration (**J. Liu** Member of Editorial Board), ‘Measurement of the total cross section and  $\rho$ -parameter from elastic scattering in pp collisions at  $\sqrt{s} = 13$  TeV with the ATLAS detector’, Eur. Phys. J. C **83** (2023) 441, arXiv:2207.12246  
▷ *On the Cover, ATLAS Physics Briefing*
- **J. Liu**, K. Dona, G. Hoshino, S. Knirek, N. Kurinsky, M. Malaker, D. W. Miller, A. Sonnenschein, M. H. Awida, P. S. Barry, K. K. Berggren, D. Bowring, G. Carosi, C. Chang, A. Chou, R. Khatiwada, S. Lewis, J. Li, S. W. Nam, O. Noroozian, T. X. Zhou, ‘Broadband Solenoidal Hodoscope for Terahertz Axion Detection’, Phys. Rev. Lett. **128** (2022) 131801, arXiv:2111.12103  
▷ *On the Cover and PRL Editors’ Suggestion, UChicago News feature*
- K. Dona, **J. Liu**, N. Kurinsky, D. Miller, P. Barry, C. Chang, A. Sonnenschein, ‘Design and performance of a multi-terahertz Fourier transform spectrometer for axion dark matter experiments’, JINST **17** (2022) P06014, arXiv:2104.07157
- ATLAS Collaboration (**J. Liu** Editor & Lead Analyzer), ‘Observation and Measurement of Forward Proton Scattering in Association with Lepton Pairs Produced via the Photon Fusion Mechanism at ATLAS’, Phys. Rev. Lett. **125** (2020) 261801, arXiv:2009.14537  
▷ *ATLAS Physics Briefing, CERN Courier feature*
- J. Amacker, W. Balunas, L. Beresford, D. Bortoletto, J. Frost, C. Issever, **J. Liu**, J. McKee, A. Micheli, S. Paredes Saenz, M. Spannowsky, B. Stanislaus, ‘Higgs self-coupling measurements using deep learning in the  $b\bar{b}b\bar{b}$  final state’, JHEP **12** (2020) 115, arXiv:2004.04240
- ATLAS Collaboration, ‘Searches for electroweak production of supersymmetric particles with compressed mass spectra in  $\sqrt{s} = 13$  TeV  $pp$  collisions with the ATLAS detector’, Phys. Rev. D **101** (2020) 052005, arXiv:1911.12606
- L. Beresford and **J. Liu**, ‘New physics and tau  $g - 2$  using LHC heavy ion collisions’, Phys. Rev. D **102** (2020) 113008, arXiv:1908.05180  
▷ *ParticleBites feature*
- I. Dawson, B. Gallop, **J. Liu**, P. Miyagawa, P. Phillips, G. Pownall, D. Robinson and A. Weidberg, ‘In situ radiation damage studies of optoelectronics in the ATLAS Semiconductor Tracker’, JINST **14** (2019) P07014
- L. Beresford and **J. Liu**, ‘Search Strategy for Stopped and Dark Matter Using the LHC as a Photon Collider’, Phys. Rev. Lett. **123** (2019) 141801, arXiv:1811.06465
- ATLAS Collaboration (**J. Liu** Editor & Lead Analyzer), ‘Search for electroweak production of supersymmetric states in scenarios with compressed mass spectra at  $\sqrt{s} = 13$  TeV with the ATLAS detector’, Phys. Rev. D **97** (2018) 052010, arXiv:1712.08119  
▷ *Two ATLAS Physics Briefings, CERN Courier feature, Oxford Physics highlight*
- A. Barr and **J. Liu**, ‘Analysing parameter space correlations of recent 13 TeV gluino and squark searches in the pMSSM’, Eur. Phys. J. C **77** (2017) 202, arXiv:1608.05379

- A. Barr and **J. Liu**, ‘First interpretation of 13 TeV supersymmetry searches in the pMSSM’, arXiv:1605.09502, Published with 1608.05379 in Eur. Phys. J. C **77** (2017) 202  
▷ *Oxford Physics highlight*

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|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| PROCEEDINGS      | • <b>J. Liu</b> on behalf of the ALICE, ATLAS, CMS, LHCb Collaborations, ‘Quantum Chromodynamics at the Large Hadron Collider’, Proceedings of the Lepton Photon Symposium 2025, arXiv:2601.01217, ATL-PHYS-PROC-2025-114 |             |
| INVITED SEMINARS | Brookhaven National Laboratory, Particle Physics Seminar, USA                                                                                                                                                             | 28 May 2026 |
|                  | University of Chicago, Enrico Fermi Institute Colloquium, USA                                                                                                                                                             | 11 May 2026 |
|                  | Rutgers University, High Energy Experimental Seminar, USA                                                                                                                                                                 | 23 Sep 2025 |
|                  | New York City College of Technology, Theoretical Physics Seminar, USA                                                                                                                                                     | 27 Mar 2025 |
|                  | New York University, CCPP Brown Bag, USA                                                                                                                                                                                  | 10 Mar 2025 |
|                  | University of Cambridge, Cavendish HEP Seminar, UK                                                                                                                                                                        | 7 May 2024  |
|                  | New York University, Experimental Particle Physics Seminar, USA                                                                                                                                                           | 5 Mar 2024  |
|                  | University of California, Irvine, Particle Physics Seminar, USA                                                                                                                                                           | 20 Feb 2024 |
|                  | Sorbonne Université, LPTHE Phenomenology Seminar, France                                                                                                                                                                  | 3 Oct 2023  |
|                  | Institut d’Astrophysique de Paris, ICAP Seminar, France (Online)                                                                                                                                                          | 22 Jun 2023 |
|                  | University of Sydney, CPPC Seminar, Australia (Online)                                                                                                                                                                    | 4 May 2023  |
|                  | Johns Hopkins University, Experimental Particle Physics Seminar, USA                                                                                                                                                      | 5 Apr 2023  |
|                  | Massachusetts Institute of Technology, LNS Colloquium, USA                                                                                                                                                                | 27 Feb 2023 |
|                  | Brown University, High Energy Experiment Seminar, USA                                                                                                                                                                     | 15 Feb 2023 |
|                  | University of Edinburgh, Experimental Particle Physics Seminar, UK                                                                                                                                                        | 27 Jan 2023 |
|                  | University of Bristol, Particle Physics Seminar, UK                                                                                                                                                                       | 7 Dec 2022  |
|                  | University of Liverpool, HEP Seminar, UK                                                                                                                                                                                  | 16 Nov 2022 |
|                  | Imperial College London, HEP Seminar, UK                                                                                                                                                                                  | 15 Jun 2022 |
|                  | University of Warwick, Particle Physics Seminar, UK                                                                                                                                                                       | 9 Jun 2022  |
|                  | Cornell University, LEPP Journal Club Seminar, USA (Online)                                                                                                                                                               | 6 May 2022  |
|                  | Perimeter Institute for Theoretical Physics, Particle Physics Seminar, Canada                                                                                                                                             | 3 May 2022  |
|                  | Case Western Reserve University, Particle/Astro Seminar, USA (Online)                                                                                                                                                     | 1 Mar 2022  |
|                  | University of Cambridge, Cavendish HEP Seminar, UK                                                                                                                                                                        | 30 Nov 2021 |
|                  | University of Tennessee, Knoxville, HEP Seminar, USA (Online)                                                                                                                                                             | 8 Sep 2021  |
|                  | Cornell University, LEPP Journal Club Seminar, USA (Online)                                                                                                                                                               | 30 Oct 2020 |
|                  | Stony Brook University, Particle Physics Seminar, USA (Online)                                                                                                                                                            | 19 Oct 2020 |
|                  | University of Pittsburgh, PITT PACC Seminar, USA (Online)                                                                                                                                                                 | 5 Aug 2020  |
|                  | SLAC Stanford University, Particle Physics Theory Seminar, USA                                                                                                                                                            | 12 Feb 2020 |
|                  | Fermilab, Cosmic Physics Center Chalk Talk, USA                                                                                                                                                                           | 14 Nov 2019 |
|                  | Fermilab, LPC Physics Forum, USA                                                                                                                                                                                          | 31 Oct 2019 |
|                  | University of Cambridge, Cavendish HEP Seminar, UK                                                                                                                                                                        | 19 Feb 2019 |
|                  | LBNL University of California, Berkeley, RPM Seminar, USA                                                                                                                                                                 | 20 Nov 2018 |
|                  | SLAC Stanford University, Joint Theory–Experiment Seminar, USA                                                                                                                                                            | 20 Apr 2018 |
|                  | Perimeter Institute for Theoretical Physics, BSM Seminar, Canada                                                                                                                                                          | 17 Apr 2018 |
|                  | University of California, Santa Cruz, SCIPP Seminar, USA                                                                                                                                                                  | 10 Apr 2018 |
|                  | University of Oxford, Dalitz Seminar, UK                                                                                                                                                                                  | 19 Jan 2017 |
|                  | University of Cambridge, Joint DAMTP–Cavendish Seminar, UK                                                                                                                                                                | 13 Jan 2017 |
| CONFERENCE TALKS | American Physical Society (APS) Global Physics Summit, Denver, USA                                                                                                                                                        | 16 Mar 2026 |
|                  | Brookhaven Forum, Brookhaven National Laboratory, USA                                                                                                                                                                     | 23 Oct 2025 |
|                  | CPAD Instrumentation Conference, University of Pennsylvania, USA                                                                                                                                                          | 8 Oct 2025  |

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|                            | QCD@LHC Conference, Stony Brook, USA                                    | 9 Sep 2025  |
|                            | Lepton Photon Symposium (Invited Plenary), Madison, Wisconsin, USA      | 26 Aug 2025 |
|                            | European Physical Society (EPS) HEP Conference, Marseille, France       | 7 Jul 2025  |
|                            | Phenomenology Symposium, Pittsburgh, USA                                | 19 May 2025 |
|                            | Cavendish HEP Research Extravaganza, Cambridge, UK                      | 6 Dec 2023  |
|                            | CADEx Collaboration Meeting, Spain (Online)                             | 16 Nov 2023 |
|                            | BREAD Collaboration Meeting, Chicago, USA                               | 5 Oct 2023  |
|                            | EPS-HEP Conference, Hamburg, Germany                                    | 24 Aug 2023 |
|                            | Dark Matter Beyond the Weak Scale Conference, Liverpool, UK             | 28 Mar 2023 |
|                            | Photon-induced Processes Workshop, Durham, UK                           | 3 Nov 2022  |
|                            | Dark Matter UK Meeting, London, UK                                      | 22 Sep 2022 |
|                            | International Conference on High Energy Physics (ICHEP), Bologna, Italy | 8 Jul 2022  |
|                            | Institute of Physics (IoP) Joint APP HEPP Conference, Harwell, UK       | 5 Apr 2022  |
|                            | Snowmass Cosmic Frontier CF2 Wavelike DM Meeting (Online)               | 8 Oct 2021  |
|                            | APS Division of Particles & Fields Meeting, Florida State, USA (Online) | 14 Jul 2021 |
|                            | Phenomenology Symposium, Pittsburgh, USA (Online)                       | 25 May 2021 |
|                            | APS April Meeting (Online)                                              | 20 Apr 2021 |
|                            | DIS Conference, Stony Brook, USA (Online)                               | 14 Apr 2021 |
|                            | LHC Forward Physics Meeting (Online)                                    | 4 Mar 2021  |
|                            | Forward Physics and QCD with LHC, EIC and Cosmic Rays (Online)          | 23 Jan 2021 |
|                            | Snowmass Energy Frontier EF09 BSM Meeting (Online)                      | 4 Sep 2020  |
|                            | ICHEP Conference, Prague, Czech Republic (Online, 2 talks)              | 30 Jul 2020 |
|                            | Snowmass Energy Frontier Joint Preparatory Session (Online)             | 8 Jul 2020  |
|                            | Higgs Couplings Conference, Oxford, UK                                  | 2 Oct 2019  |
|                            | Young Experimentalists & Theorists Institute, Durham, UK                | 8 Jan 2019  |
|                            | LHC Forward Physics Workshop, CERN, Switzerland                         | 18 Dec 2018 |
|                            | Oxford–Durham Di-Higgs Workshop, Durham, UK                             | 6 Nov 2018  |
|                            | SUSY Conference, Barcelona, Spain                                       | 23 Jul 2018 |
|                            | ICHEP Conference (Plenary prize talk), Seoul, South Korea               | 6 Jul 2018  |
|                            | DM@LHC Workshop, Heidelberg, Germany                                    | 5 Apr 2018  |
|                            | Institute of Physics Joint APP HEPP Conference, Bristol, UK             | 27 Mar 2018 |
|                            | Young Theorists Forum, Durham, UK                                       | 10 Jan 2018 |
|                            | Young Theorists Forum, Durham, UK                                       | 11 Jan 2017 |
|                            | 2nd LHC BSM Reinterpretation Workshop, CERN, Switzerland                | 14 Dec 2016 |
|                            | Theoretical Particle Physics Summer School, Manchester, UK              | 31 Aug 2016 |
|                            | 1st LHC BSM Reinterpretation Workshop, CERN, Switzerland                | 17 Jun 2016 |
| COLLABORATION<br>PLENARIES | ATLAS SM Group Plenary, CERN, Switzerland                               | 5 Mar 2026  |
|                            | ATLAS SM Soft QCD Plenary, CERN, Switzerland                            | 1 Dec 2025  |
|                            | ATLAS US Summer Workshop, Ann Arbor, Michigan, USA                      | 23 Jul 2025 |
|                            | ATLAS SM Electroweak Plenary, CERN, Switzerland                         | 4 Oct 2024  |
|                            | ATLAS UK SM+BPHYS Meeting, Oxford, UK                                   | 2 Oct 2024  |
|                            | ATLAS SM Workshop, Ljubljana, Slovenia                                  | 3 Sep 2024  |
|                            | ATLAS Week Plenary on ITk Strips, Thessaloniki, Greece                  | 20 Jun 2024 |
|                            | ATLAS HV Breakdown Task Force Meeting (Online)                          | 16 Feb 2024 |
|                            | ATLAS SM Workshop, Prague, Czechia                                      | 12 Sep 2023 |
|                            | ATLAS HV Breakdown Task Force Meeting (Online)                          | 28 Jul 2023 |
|                            | ATLAS ITk Strips Module Meeting (Online)                                | 18 Jul 2023 |
|                            | ATLAS UK Meeting, Manchester, UK                                        | 9 Jan 2023  |

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|                         | ATLAS SUSY Workshop, CERN, Switzerland                                 | 19 Sep 2022       |
|                         | ATLAS ITk WP12 Meeting, Birmingham, UK                                 | 31 Mar 2022       |
|                         | ATLAS Upgrade Physics (Online)                                         | 22 Apr 2021       |
|                         | ATLAS SM Plenary (Online)                                              | 15 Apr 2021       |
|                         | ATLAS Run 4 Roman Pot Meeting (Online)                                 | 26 Mar 2021       |
|                         | ATLAS Idea Day (Online)                                                | 20 Jan 2021       |
|                         | ATLAS Roman Pot General Meeting (Online)                               | 7 Oct 2020        |
|                         | ATLAS SM Jamboree (Online)                                             | 30 Sep 2020       |
|                         | ATLAS SM Plenary Analysis Closure (Online)                             | 2 Jul 2020        |
|                         | ATLAS Week Plenary on SM & AFP (Online)                                | 24 Jun 2020       |
|                         | ATLAS Roman Pot General Meeting, CERN, Switzerland                     | 23 Jan 2020       |
|                         | ATLAS Hadronic Final State Forum, Chicago, USA                         | 20 Dec 2019       |
|                         | ATLAS UK SUSY–Exotics Meeting, Cambridge, UK                           | 11 Apr 2019       |
|                         | ATLAS SUSY Run 3 Ideas, CERN, Switzerland                              | 22 Nov 2018       |
|                         | ATLAS SUSY Plenary, CERN, Switzerland                                  | 19 Jul 2018       |
|                         | ATLAS Exotics Workshop, Rome, Italy (poster)                           | 29 May 2018       |
|                         | ATLAS Week Plenary on SUSY, CERN, Switzerland                          | 21 Feb 2018       |
|                         | ATLAS Analysis Open Presentation, CERN, Switzerland                    | 1 Nov 2017        |
|                         | ATLAS Exotics–SUSY Workshop, Bucharest, Romania                        | 12 May 2017       |
|                         | ATLAS SUSY Electroweak, CERN, Switzerland                              | 10 Feb 2017       |
|                         | ATLAS UK Meeting, Liverpool, UK                                        | 5 Jan 2017        |
|                         | ATLAS Week Plenary on SCT/ID Gen, CERN, Switzerland                    | 17 Oct 2016       |
| MENTORING<br>& TEACHING | Postdoctoral researcher mentoring                                      |                   |
|                         | Sebastian Rutherford Colmenares, NYU                                   | 2025 – present    |
|                         | Graduate student mentoring                                             |                   |
|                         | Amelia Stevens, research assistant, NYU                                | 2025 – present    |
|                         | Anna Mullin, PhD student, Cambridge                                    | 2022              |
|                         | Kristin Dona, PhD student, UChicago                                    | 2019 – 2021       |
|                         | Savannah Clawson, PhD student, Manchester                              | 2019 – 2020       |
|                         | Santiago Paredes Saenz, PhD student, Oxford                            | 2018 – 2020       |
|                         | Beojan Stanislaus, PhD student, Oxford                                 | 2018 – 2020       |
|                         | Gabriel Gallardo, PhD student, Oxford                                  | 2018 – 2019       |
|                         | Gavin Pownall, PhD student, Oxford                                     | 2018 – 2019       |
|                         | Undergraduate student mentoring                                        |                   |
|                         | Liliana Valle, REU summer student, UChicago                            | 2021              |
|                         | Joe Ray, MPhys student, Oxford                                         | 2018 – 2019       |
|                         | Dan Kreso, MPhys student, Oxford                                       | 2017 – 2018       |
|                         | Alessandro Micheli, summer student, Oxford                             | 2018              |
|                         | Ewan McCullough, summer student, Oxford                                | 2017              |
|                         | Charles Sillett, summer student, Oxford/CERN                           | 2017              |
|                         | New York University courses taught                                     |                   |
|                         | PHYS-GA 2027 Graduate Particle Physics                                 | Spring 2026, 2025 |
|                         | Member of PhD Thesis Defense Committees, NYU Physics                   |                   |
|                         | Zubair Bhatti, Matthew Drnevich                                        | 2026              |
|                         | Member of PhD Oral Qualification Exam (1 student), NYU Physics         | 2026              |
|                         | Panelist, ‘Working with your PhD supervisor’ workshop, Trinity College | 2023              |

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|                         | Trinity College Postdoc-PhD mentoring programme                                               | 2022               |
|                         | Lecturer, Oxford undergraduate summer students: ‘The Standard Model’                          | Jul 2018           |
|                         | Tutor, Oxford 3rd year undergraduates: ‘Flows, Fluctuations & Complexity’                     | May 2016           |
|                         | Tutor, Oxford 3rd year undergraduates: ‘Subatomic Physics’                                    | May 2016           |
| LEADERSHIP<br>& SERVICE | Convener of QCD physics sessions, LHCP Conference, Paris, France                              | 2025 – 2026        |
|                         | Convener of QCD physics sessions, EPS-HEP Conference, Marseille, France                       | 2025               |
|                         | ▷ Conference abstract reviewer, co-organizer of agenda, and chair for parallel talk sessions  |                    |
|                         | Convener for LHC Physics Centre at CERN (LPCC) Forward Physics Group                          | 2023 – 2025        |
|                         | ▷ ATLAS contact, co-organizer, and convener for March, July and December 2024 workshops       |                    |
|                         | Cambridge ATLAS Inner Tracker (ITk) strips upgrade module production                          | 2021 – 2024        |
|                         | ▷ Qualified cleanroom site for module production, mechanical and electrical quality control   |                    |
|                         | ▷ Led assembly and quality control of 60+ ITk strips modules to prove production readiness    |                    |
|                         | ▷ Module thermal cycling: pioneered electromechanical stress analysis and interposer solution |                    |
|                         | Convener of ATLAS Standard Model Soft QCD physics group                                       | 2022 – 2024        |
|                         | ▷ Coordination role organizing program of 20 analyses probing non-perturbative dynamics       |                    |
|                         | ▷ Chaired bi-weekly plenary meetings, steered physics group strategy and workshop agendas     |                    |
|                         | Convener for Run 4 AFP Task Force on HL-LHC physics projections                               | 2021 – 2022        |
|                         | ▷ Coordination role organizing physics and performance for ATLAS Forward Proton upgrade       |                    |
|                         | ▷ Chaired meetings discussing beam optics, detector performance and physics opportunities     |                    |
|                         | ATLAS Forward Proton combined performance and data preparation                                | 2019 – 2020        |
|                         | ▷ Led scrutiny of AFP data quality, detector efficiency, energy scale calibration, simulation |                    |
|                         | ▷ Delivered first collaboration-wide AFP performance recommendations for 2017 dataset         |                    |
|                         | ITk Strip Module Level 3 Manager (Deputy), US ATLAS Upgrade                                   | 2026 – present     |
|                         | Contact Editor, ATLAS Standard Model proton-oxygen analysis 2604.05512                        | 2025 – present     |
|                         | Institute Representative (Team Leader), NYU ATLAS                                             | 2025 – present     |
|                         | Computer and Departmental Webpage Committee, NYU Physics                                      | 2025 – present     |
|                         | Editorial Board member for ATLAS SM Soft QCD analysis                                         | 2025 – present     |
|                         | Analysis Contact, ATLAS Standard Model photon fusion production in Run 3                      | 2024 – present     |
|                         | Graduate physics open house talk, NYU Department of Physics                                   | 2026, 2025         |
|                         | ATLAS paper review (Institute Reading), Standard Model, NYU ATLAS                             | 2025               |
|                         | Invited member-at-large for subgroup convenors, ATLAS Standard Model                          | 2024               |
|                         | Grant expert reviewer for National Science Centre, Poland                                     | 2023               |
|                         | Journal referee for Annalen der Physik                                                        | 2023               |
|                         | ATLAS paper review (Institute Reading), Heavy-ion and Higgs, Cambridge ATLAS                  | 2023               |
|                         | Contributing author, Snowmass whitepaper ‘Axion Dark Matter’, 2203.14923                      | 2022               |
|                         | UChicago ATLAS website redesign, authored recruitment and mission statements                  | 2021               |
|                         | Journal referee for Nuclear Physics B                                                         | 2021               |
|                         | ATLAS paper review (Institute Reading), Trigger, UChicago ATLAS                               | Dec 2019, Sep 2020 |
|                         | Coordinator for Snowmass Letter of Interest on terahertz axions                               | Aug 2020           |
|                         | Grainger Fellowship website development, UChicago Physics                                     | Jul 2020           |
|                         | Postdoctoral community working group, UChicago KICP                                           | Jul 2020           |
|                         | Journal club, UChicago ATLAS group                                                            | May 2020           |
|                         | Graduate open day speaker & poster, Oxford Particle Physics                                   | Dec 2018           |
|                         | ATLAS paper review (Institute Reading), Supersymmetry, Oxford ATLAS                           | Jul 2018           |
|                         | Organising committee member, YETI 2017, Durham University                                     | Sep 2016           |
|                         | Physics aptitude test marker, Oxford undergraduate admissions                                 | Nov 2015           |

|          |                                                                                 |                |
|----------|---------------------------------------------------------------------------------|----------------|
|          | Resident advisor for US visiting students, Williams-Exeter Programme at Oxford  | 2012 – 2014    |
|          | Publicity officer, Oxford Undergraduate Physics Conference                      | Mar 2013       |
| OUTREACH | ATLAS Physics Briefing, proton-oxygen cosmic-ray news with ATLAS Outreach       | 15 Apr 2026    |
|          | NYU EPP News, wrote news article on proton-oxygen cosmic-ray result             | 10 Apr 2026    |
|          | Faculty Science Communication Workshop, NYU Carter Journalism Institute         | Oct–Nov 2025   |
|          | First-gen panelist, NYU Conference for Undergraduate Women in Physics           | 25 Jan 2025    |
|          | The Conversation, news article with science journalists on taus in heavy ions   | 17 Oct 2023    |
|          | Sidney Sussex MCR Research Conference, Cambridge, invited guest speaker         | 4 Mar 2023     |
|          | Interviewed for UChicago news article on BREAD axion experiment proposal        | 12 Apr 2022    |
|          | ATLAS Physics Briefing, news article on taus in heavy ions result               | 7 Apr 2022     |
|          | Trinity College Science Society Symposium, ‘Dark matter and colossal colliders’ | 13 Mar 2022    |
|          | CERN Courier magazine article, ‘The LHC as a photon collider’                   | 15 Aug 2020    |
|          | ATLAS Physics Briefing, online news article on AFP result                       | 30 Jul 2020    |
|          | Trinity Forum, Cambridge, ‘Dark matter at the LHC’ invited talk                 | 5 Jun 2019     |
|          | Pint of Science, Manchester, ‘Using light to reveal the darkness’ invited talk  | 20 May 2019    |
|          | Oxford HiggsHunters, mentoring secondary school students                        | 26–27 Jun 2018 |
|          | CERN school visit, hosting 35 sixth form students, teacher engagement           | 15 Feb 2018    |
|          | FQXi essay, information in physics and biology                                  | 3 Mar 2017     |
|          | Lunch with CERN scientists, physics discussions with school students            | 21 Oct 2016    |
|          | Oxfordshire Science Festival, all-day live physics demos on Broad Street        | 26 Jun 2016    |
|          | Perimeter Institute PSI Blog, ‘Surprises of spin’                               | 15 May 2016    |
|          | Out of this World: Family Fun Day, targeting under-represented at university    | 17 Apr 2016    |
|          | Oxford Catz Exchange, ‘Seeing the unseeable’ talk                               | 25 Feb 2016    |
|          | Oxford FameLab, ‘Listening to numbers’ talk                                     | 25 Jan 2016    |
|          | Oxford Stargazing Live, all-day live demos at particle physics stall            | 16 Jan 2016    |
|          | Oxford Science Christmas Lectures, mentoring school students                    | 8 Dec 2015     |
|          | Oxford Flash Talk Physics, ‘Fiddling with physics’ invited talk                 | 4 Dec 2013     |
|          | Institute of Physics essay, ‘Predicting the unpredictable’                      | 30 May 2012    |